



# Financial Inclusion in Africa

Predicting Financial Inclusion using Machine Learning

# Agenda



1. Project Goal
2. Dataset
3. Data Exploration (EDA)
4. Data Cleaning & Preprocessing
5. Feature Engineering

6. Machine Learning Models
7. Model Evaluation
8. Feature Importance
9. Conclusion





# Project Goal

Financial inclusion in Africa  
Predict financial service usage  
Identify underserved populations  
Support data-driven decisions  
Improve access to financial services



# Dataset Structure



Zindi competition dataset

Train / Test / Submission

Demographic information

Financial behaviour

Categorical + numerical features

## Dataset Structure

**Train.csv**

**23,524 rows · 13 columns**

**Test.csv**

**10,086 rows · 12 columns**

**VariableDefinitions.csv**

**Feature descriptions**

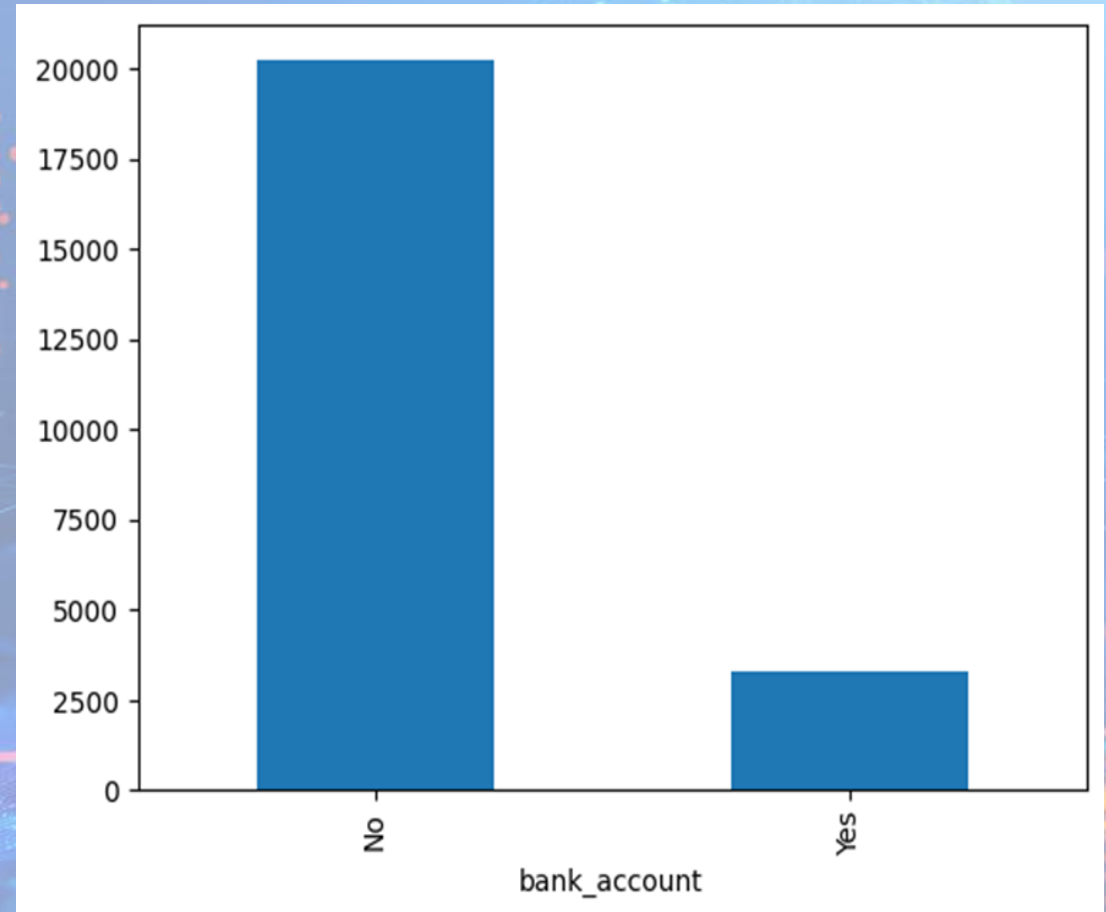
# Target variable

bank\_account

No → 85.9 %

Yes → 14.1 %

Imbalanced classification





# Exploratory Data Analysis

Cellphone access

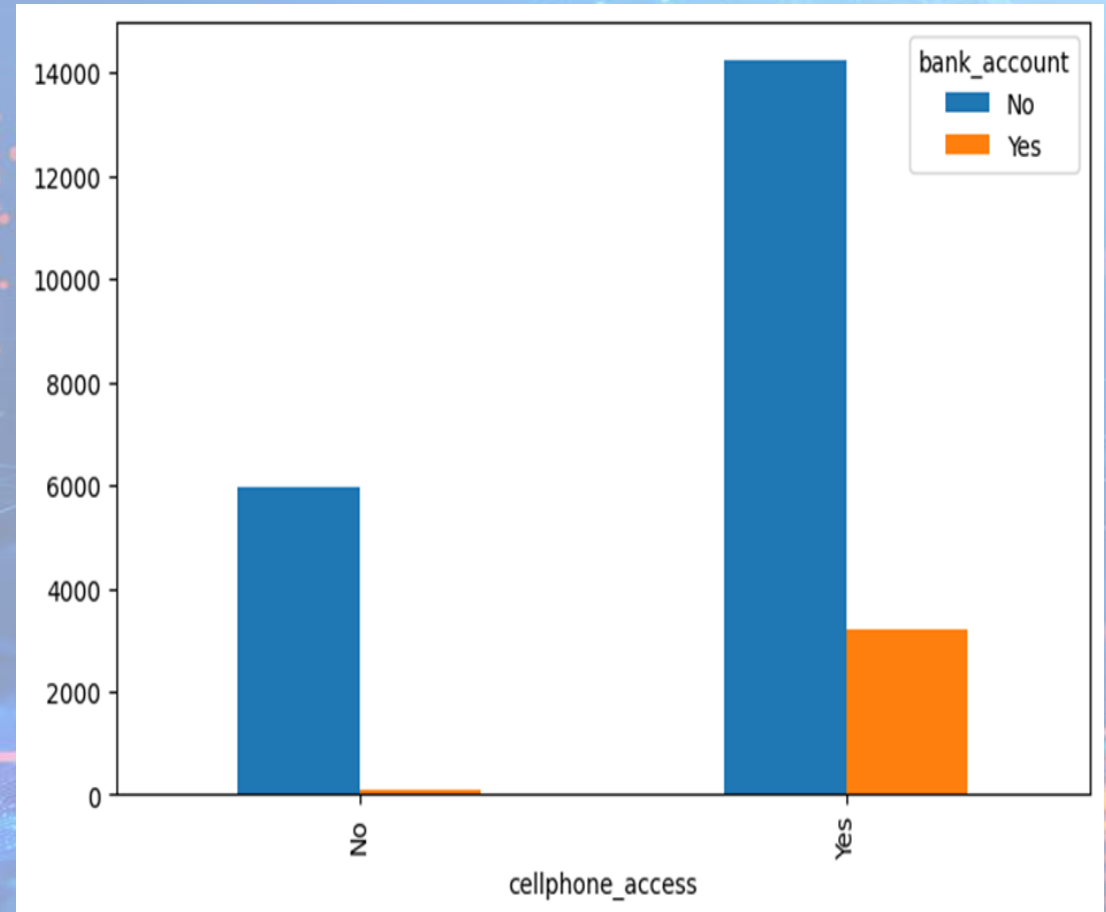
Education level

Location type

Country differences

Age distribution

Household size



# EDA



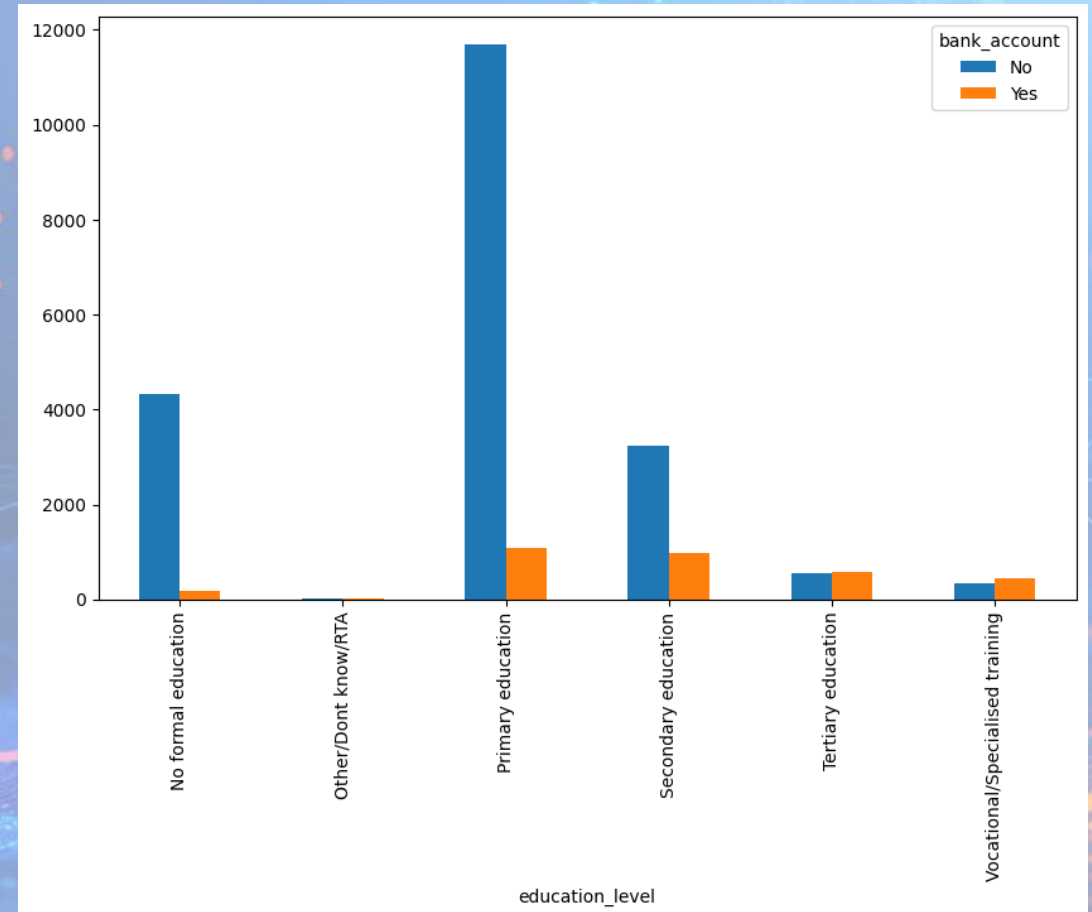
Data understanding

Missing values

Feature distributions

Class imbalance

Initial correlations



# Machine Learning



Missing value handling

Categorical encoding

Feature selection

Train / Test split

Scaled input features

**Raw Data**



**Preprocessing  
(One-Hot Encoding)**



**Dummy**

**Logistic Regression**

**Decision Tree**

**Random Forest**



**Model Evaluation**



# Model Comparison



Logistic Regression

Decision Tree

Random Forest

XGBoost

Performance evaluation

|   | model               | accuracy | mae      | recall_class_1 | precision_class_1 | f1_class_1 |
|---|---------------------|----------|----------|----------------|-------------------|------------|
| 0 | Dummy Baseline      | 0.859299 | 0.140701 | 0.00           | 0.00              | 0.00       |
| 1 | Logistic Regression | 0.790648 | 0.209352 | 0.76           | 0.38              | 0.51       |
| 2 | Decision Tree       | 0.698193 | 0.301807 | 0.83           | 0.30              | 0.44       |
| 3 | Random Forest       | 0.854410 | 0.145590 | 0.45           | 0.48              | 0.46       |

# Feature Importance



Feature importance analysis

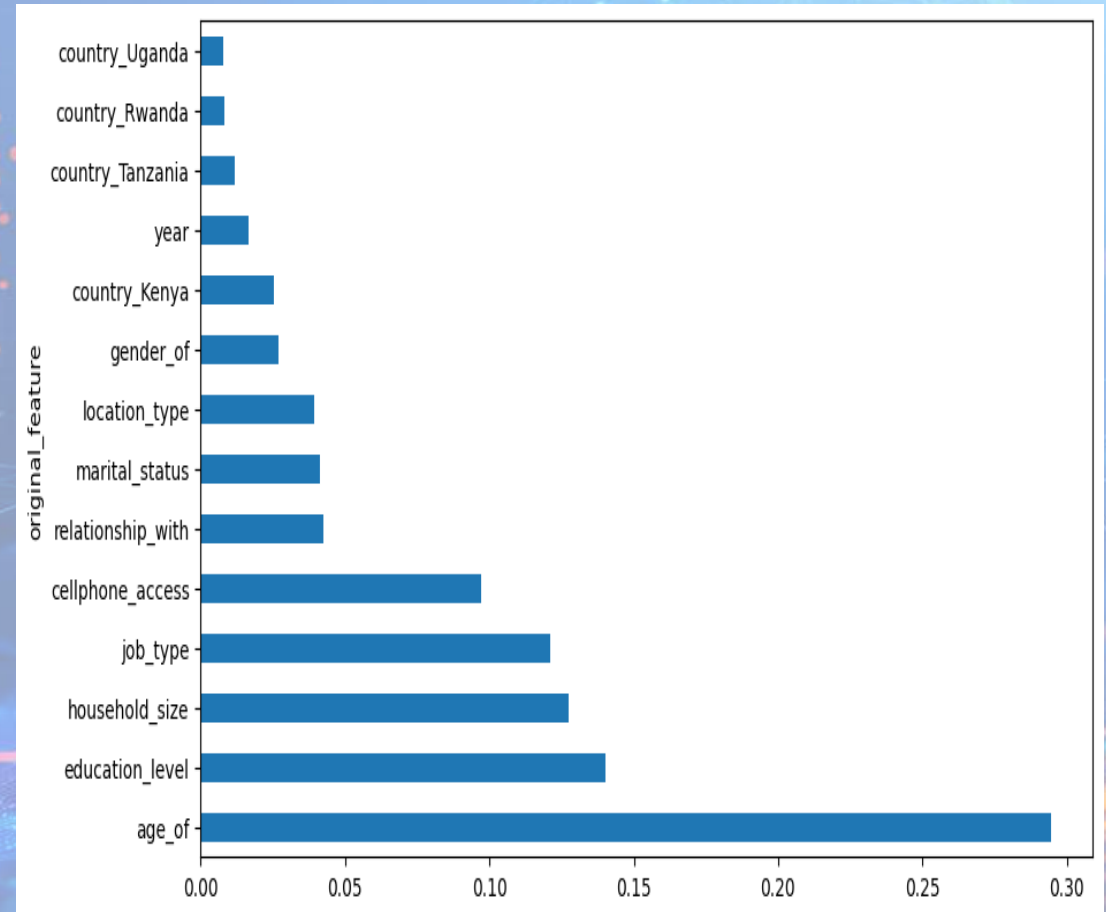
Household location most influential

Wealth indicators improve prediction

Demographic features contribute

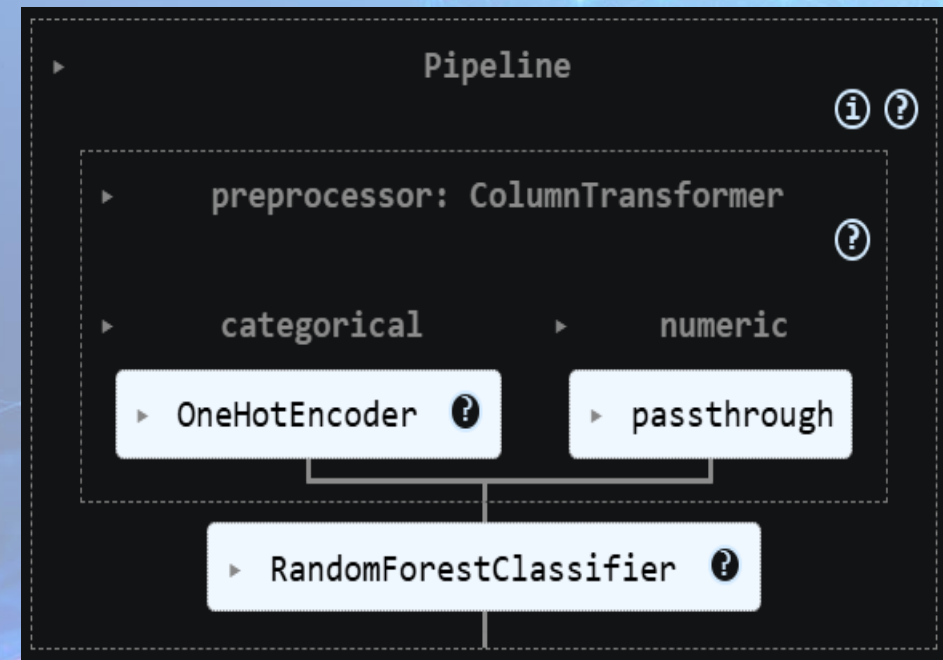
Less relevant variables removed

Better interpretability



# Final Model

Random Forest selected  
Hyperparameter tuning applied  
Cross-validation performed  
Stable generalization  
Improved prediction quality  
Robust final model





# Conclusion



Random Forest selected

Best prediction performance

Robust cross-validation

Real-world application potential

More socioeconomic features

Advanced ensemble models

Further model optimization

Probability-based predictions